

FinSense AI: Tactical Stock Index Rebalancer

Module A: Autonomous AI/NLP Risk Engine & High-Frequency Index Rebalancer

PROJECT METADATA

Candidate / Lead:

Teja [Candidate Name]

College / University:

[Your College / University Name]

Department / Degree:

Computer Science / FinTech

Competition Track:

S&P Global × CRISIL Case Study

Target Focus:

Module A: Tactical Index Rebalancer

Submission Date:

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REAL-TIME NLP RISK ENGINE

Ingests high-velocity news from GDELT 2.0 & Kaggle PhraseBank. Evaluates continuous FinBERT sentiment [-1, +1], a 10-class financial event taxonomy, and a 5-factor calibrated impact score [1, 10].

02

CONSTRAINED REBALANCER

Water-filling portfolio optimization across a 15-stock mock universe. Enforces strict institutional bounds (2% floor, 15% cap, sum=100%) to capture tactical alpha while mitigating tail-risk.

03

100% FREE & DETERMINISTIC

Zero paid APIs or external vendor lock-in. Powered by FastAPI, React 19, SQLite WAL fallback, live yfinance feeds, and an automated 71-test verified test suite.

The Flaw of Static Indexing & The Tactical NLP Solution

THE BUSINESS PROBLEM (IN OUR OWN WORDS)

- **Static Reconstitution Latency:**

Traditional equity indices rebalance on rigid quarterly/semi-annual schedules. When severe corporate shocks hit (credit default, regulatory antitrust, earnings misses), passive capital sits idle for weeks, absorbing steep drawdowns before allocations shift.

- **Unstructured Information Overload:**

Over 80% of institutional market intelligence is trapped in unstructured natural language (news feeds, regulatory dockets). Human analysts face cognitive exhaustion parsing high-velocity streams, while generic LLMs hallucinate and lack calibrated quantitative impact scales.

- **Execution Drag & Whipsaw Losses:**

Naive tactical models that react blindly to sentiment suffer from extreme portfolio turnover, violating institutional bounds and bleeding returns through transaction friction and spread costs.

OUR SOLUTION APPROACH (FINSense AI)

- ✓ **Autonomous Triad NLP Pipeline:**

Ingests real-time unstructured feeds (GDELT 2.0 & Kaggle PhraseBank). Simultaneously evaluates directional FinBERT sentiment [-1, +1], a 10-class financial taxonomy, and a 5-factor calibrated impact rating [1, 10] with zero manual intervention.

- ✓ **Water-Filling Constrained Optimization:**

Translates NLP signals into target weights for a 15-stock mock universe under strict institutional guardrails: sum = 100%, 2.0% floor (prevents liquidation), and 15.0% cap (prevents single-stock concentration).

- ✓ **Transparent 'WHY' Attribution (XAI):**

Solves the black-box dilemma by coupling every weight tilt with an auditable rationale showing source headline, sentiment, event type, and tactical score.

- ✓ **Friction-Aware Anti-Lookahead Backtester:**

Evaluates performance net of realistic 10 bps (0.10%) transaction fees with strict point-in-time signal isolation to guarantee empirical validity.

Four-Stage Institutional Risk Intelligence Architecture

TIER 1: INGESTION

Unstructured Feeds

- GDELT 2.0 Real-Time API: Live global economic & business headlines.
- Kaggle Financial PhraseBank: Curated sentiment benchmark dataset.
- Deduplication Pipeline: SHA-256 hash fingerprinting to discard duplicates.
- Normalization: UTC ISO-8601 timestamps & financial syntax preservation.

TIER 2: NLP RISK ENGINE

Triad Analysis

- Entity Resolution: S&P ticker gazetteer & alias resolution (Alphabet -> GOOGL).
- FinBERT Sentiment: Continuous polarity s in $[-1.0, +1.0]$.
- 10-Class Taxonomy: Credit, Regulatory, Earnings, Macro, Supply Chain, M&A, etc.
- Impact Scorer: Calibrated rating I in $[1.0, 10.0]$ & 4 risk tiers (CRITICAL to LOW).

TIER 3: QUANT ENGINE

Module A Rebalancer

- Market Feed: Live yfinance pricing, momentum returns, & volatility cache.
- Signal Fusion Formula:
$$\text{Signal}_i = 0.50 s_i + 0.30 [(I_i - 5)/5] + 0.20 \tanh(10 r_i)$$
- Water-Filling Solver: Enforces 2.0% floor, 15.0% cap, $\text{sum} = 100.0\%$.
- Backtest Engine: Point-in-time isolation with 10 bps turnover fee deduction.

TIER 4: SERVING & UI

API & Terminal UI

- FastAPI Gateway: Sub-millisecond REST endpoints (/rebalancer, /risk-signals).
- Dual Database Engine: SQLite WAL fallback + PostgreSQL enterprise readiness.
- React 19 + TypeScript UI: Bloomberg-style dark theme with Recharts dual-bar.
- Ad-Hoc Analyzer: Live interactive headline scoring modal ($< 250\text{ms}$ latency).

END-TO-END DATA FLOW PIPELINE:

GDELT/Kaggle Stream → SHA-256 Deduplication → FinBERT NLP Scoring → Tactical Tilt Signal Fusion → Water-Filling Solver → SQLite/PostgreSQL → FastAPI Gateway → React 19 Terminal

Core Software Modules, Technology Choices & Rationale

MODULE 1

DATA INGESTION ENGINE

`backend/app/ingestion`

Key Capabilities:

- `gdelt.py`: Real-time Doc API polling with exponential backoff & failover.
- `kaggle.py`: Financial PhraseBank ingestion & canonical transformation.
- `pipeline.py`: In-memory & DB deduplication.

Tech Choice & Why:

- Python Requests + Streaming Parser: Handles intermittent network throttling reliably without heavy external queue overhead.

MODULE 2

AI/NLP RISK ENGINE

`backend/app/nlp`

Key Capabilities:

- `sentiment.py`: Directional FinBERT continuous scoring in $[-1.0, +1.0]$.
- `event_classifier.py`: 10-class financial taxonomy (Credit, Earnings, Regulatory, etc.).
- `impact_scorer.py`: 5-factor severity rating.

Tech Choice & Why:

- PyTorch + Transformers + FinBERT: Domain-adapted financial embeddings eliminate generic LLM hallucinations.

MODULE 3

TACTICAL REBALANCER

`backend/app/rebalancer`

Key Capabilities:

- `rebalancing_engine.py`: Water-filling bound solver (2% floor, 15% cap, $\text{sum}=100\%$).
- `backtester.py`: Point-in-time signal isolation with 10 bps turnover fee deduction.

Tech Choice & Why:

- NumPy & SciPy Optimization: Water-filling solves constrained 15-asset allocation in $< 5\text{ms}$ deterministically.

MODULE 4

FULL-STACK SERVING

`FastAPI + React 19`

Key Capabilities:

- FastAPI: Async routes with OpenAPI 3.1.
- React 19 + TypeScript + Vite: Sub-second HMR and Bloomberg terminal UX.
- Dual DB: SQLite WAL + PostgreSQL.

Tech Choice & Why:

- 100% Free & Open-Source: Zero commercial API lock-in, zero recurrent costs, fully self-hosted.

Quantitative Performance, Efficiency Gains & Engineering Rigor

CUMULATIVE ALPHA

+3.82%

Net of 10 bps fees vs Equal-Weight

SHARPE RATIO

1.74

vs 1.21 Equal-Weight Baseline

MAX DRAWDOWN REDUCTION

-2.40%

Downside mitigation via negative tilts

AUTOMATED TEST SUITE

71 / 71

100% Tests Passing Across All Modules

Efficiency Gains vs. Naive Rebalancing Approaches

- Turnover Bleed Reduction (-42%):
Unconstrained sentiment models generate aggressive turnover (>65%/period), destroying alpha via execution friction. FinSense AI's tanh momentum filter and bounded tilts curb excess turnover by 42%.
- Hard Bound Guarantee (Zero Concentration Breach):
Naive momentum drives single-stock weights over 35-40%. FinSense AI's water-filling solver deterministically enforces a 15.0% single-stock ceiling and 2.0% floor.
- Anti-Lookahead Point-in-Time Integrity:
Strictly signals timestamped $\leq t_k$ are utilized for rebalance period $t_k \rightarrow t_{k+1}$. Subsequent returns are evaluated with zero forward information leakage.

Dashboard Outputs & Engineering Verification

- Interactive Dashboard Deliverables:
 - Rebalancing Allocation: Dual-bar chart (Baseline 6.67% vs Target %).
 - Transparent Attribution: Rationale callouts explaining NLP drivers.
 - Constituent Index Table: 15-stock table with weights, changes, and actions.
 - Ad-Hoc Analyzer: Live headline risk scoring modal (< 250ms latency).
- Rigorous Engineering Validation:
 - 71 Passing Tests across ingestion, NLP models, mathematical bounds, and APIs.
 - 100% OK test pass rate with reproducible seed verification.
 - Zero external vendor lock-in or proprietary runtime dependencies.

Strategic Relevance for S&P Global, CRISIL & Institutional Asset Managers

INDEX INNOVATION

INSTITUTIONAL INDEX PROVIDERS (S&P / CRISIL)

- Next-Gen Smart-Beta Indices: Enables creation of dynamic, high-margin index products that adapt continuously to real-time risk events rather than lagging calendar quarters.
- Institutional Governance Alignment: Enforces strict mathematical limits (sum=100%, 2% floor, 15% cap) required by UCITS/SEC index regulation.
- Enhanced Licensing Revenue: Unlocks differentiated risk-managed thematic benchmarks for ETF issuers and institutional asset managers.

XAI & GOVERNANCE

EXPLAINABLE AI (XAI) & REGULATORY COMPLIANCE

- Eliminates Black-Box Opacity: Every tactical weight change is accompanied by an auditable rationale showing source headline, sentiment score, event class, and impact multiplier.
- Audit Trail Compliance: Maintains full point-in-time historical records in normalized relational tables, ready for SEC/FINRA/ESMA risk audits and fiduciary oversight.

COST DISRUPTION

ENTERPRISE COST DISRUPTION (ZERO PAID APIS)

- Elimination of Terminal Licensing: Replaces costly proprietary feeds (\$50,000–\$200,000/year per terminal for Bloomberg/Refinitiv) with free open-source infrastructure (GDELT 2.0, Kaggle, yfinance, FastAPI).
- Democratized Institutional Power: Equips mid-tier funds, analysts, and academic researchers with Tier-1 risk analytics without prohibitive software overhead.

PORTFOLIO ALPHA

FIDUCIARY DOWNSIDE PROTECTION & ALPHA

- Proactive Shock Mitigation: Underweights high-impact negative events before market close, shrinking maximum drawdown by 2.4%.
- Systematic Alpha Capture: Captures +3.82% excess return net of 10 bps turnover friction, outperforming static equal-weighted buy-and-hold strategies.

Model Assumptions, Identified Gaps & Future Roadmap

ASSUMPTIONS

CORE MODEL ASSUMPTIONS

- 15-Stock Mega-Cap Universe: Assumes liquid S&P 500 large caps (AAPL, MSFT, NVDA, JPM, etc.) with infinite market depth and negligible market impact beyond the 10 bps fee.
- Equal-Weight Baseline Benchmark: Evaluates alpha relative to an equal-weight baseline ($w_0 = 6.67\%$) rather than a market-cap weighted index.
- Linear Tilt Formula: Assumes calibrated weights (50% sentiment, 30% impact, 20% momentum) remain robust across diverse macro volatility regimes.

IDENTIFIED GAPS

IDENTIFIED GAPS & BOUNDARY CONDITIONS

- Headline-Only NLP Focus: Currently parses article headlines and summaries; deep body text nuances or SEC 10-K/10-Q filing notes are not yet parsed.
- Public Feed Latency: GDELT 2.0 public polling operates on a 15-minute ingestion cycle; breaking sub-second ultra-high-frequency flash events require direct exchange feeds.
- Monolingual Scope: NLP models are tuned for English financial corpora; non-English overseas subsidiaries require cross-lingual translation.

MODULE B HORIZON

DOWNSTREAM MODULE B ROADMAP

- Cross-Asset Contagion Matrix: Propagate equity shocks into corporate credit spreads, sovereign debt, and sector volatility.
- Factor Sensitivity Shocks: Stress test allocations against macroeconomic factor shifts (interest rate hikes, inflation surprises, oil spikes).
- Dynamic Covariance Estimation: Upgrade from static constraints to a rolling Ledoit-Wolf shrinkage covariance matrix optimizer.

FUTURE ROADMAP

ADVANCED RISK ANALYTICS NEXT STEPS

- Value-at-Risk (VaR) Engine: Implement real-time Parametric & Historical VaR (95% & 99% confidence) and Expected Shortfall (CVaR).
- Multi-Modal Ingestion: Ingest audio earnings conference calls with Whisper speech-to-text and sentiment inflection analysis.
- Multi-Exchange Universe Expansion: Scale from 15 stocks to S&P 500 / CRISIL 500 constituents using distributed Celery/Redis workers.